

YUZE ZHAO

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Education

- Beijing Normal University (BNU)** Beijing, China
• *M. S. - Pure Mathematics* Sep. 2023 – Now
Advisors: Prof. Dachun Yang and Prof. Wen Yuan
- Dalian University of Technology (DUT)** Dalian, China
• *B. S. - Mathematics and Applied Mathematics* Sep. 2019 – Jul. 2023
Advisors: Prof. Junfeng Li and Prof. Suqing Wu

Publications & Working Papers

- Y. Chen, D. Yang, W. Yuan and **Y. Zhao**, **Embeddings and the φ -transform characterization of Besov–Morrey Spaces**, in preparation.
- F. Bu, D. Yang, W. Yuan and **Y. Zhao**, **Matrix Weights, Maximal Operators, Calderón–Zygmund Operators, and Besov–Triebel–Lizorkin-Type Spaces — A Survey**, Anal. Theory Appl. (to appear), 2025.
- S. Wu and **Y. Zhao**, **Truncations of Logarithmic Besov Spaces**, finished and currently under revision.

Research Interests

Harmonica Analysis, Matrix Sensing, Optimization.

Research Experience

- Matrix-Weighted Besov–Morrey spaces** Oct. 2025 – Now
• *Advisor:* Dachun Yang and Wen Yuan
 - Matrix-Weighted Besov–Morrey Spaces:** established the φ -transform characterizations of matrix-weighted Besov–Morrey spaces; collected characterizations of both scalar and matrix weights.
 - Embeddings of Matrix-Weighted Besov–Morrey Spaces:** investigated the embeddings from matrix-weighted Besov–Morrey to Besov-type spaces via the φ -transform characterizations.
- Qualitative and Quantitative Matrix-Weighted inequalities** Jan. 2023 – Oct. 2025
• *Advisor:* Dachun Yang and Wen Yuan
 - Matrix Weights:** established new characterizations of A_∞ -matrix weights; collect characterizations of both scalar and matrix weights.
 - Maximal Operators:** surveyed the maximal operators associated with matrix weights as well as weighted-norm inequalities.
 - Fractional Integral Operators and Calderón–Zygmund Operators:** investigated the boundedness of fractional integral operators and Calderón–Zygmund operators.
 - Dimensions of Weights:** recalled the concepts of dimensions of weights; proposed some open question.
- Trucations of Logarithmic Besov Spaces** Sept. 2022 – Jun. 2023
• *Advisor:* Junfeng Li and Suqing Wu
 - Interpolation:** established the interpolation of logarithmic Besov spaces with $p \in (0, 1)$.
 - Boundedness of the Truncation Operator on Besov Spaces:** established the boundedness on the Besov space of the truncation operator.
- Quantitative strategy research with artificial intelligence** Mar. 2021 – Apr. 2022
• *Advisor:* Mingsong Cheng and Ben Duan
 - The Undergraduate Student Innovation Practice Program:**
 - (i) studied key machine learning algorithms (decision trees and SVM);
 - (ii) developed and implemented a dual moving average trading strategy;
 - (iii) backtested the strategy on historical market data, achieving relatively stable returns.

Awards & Honors

• National Scholarship, Beijing Normal University	2025
• Outstanding TA for Undergraduate Courses, Beijing Normal University	2024
• 1st Prize Academic Scholarship, Beijing Normal University	2024
• 1st Prize Freshman Scholarship, Beijing Normal University	2023
• Outstanding Undergraduate Thesis Award, Dalian University of Technology	2023
• Hua Luogeng Scholarship, Chinese Academy of Sciences	2022
• Excellent Campers of Summer Camp, Shanghai Jiao Tong University	2022
• Excellent Campers of Summer Camp, Beijing Normal University	2022
• 3rd Prize of Undergraduate Mathematical Contest in Modeling, Liao Ning Province	2021
• 3rd Prize of Chinese Mathematics Competitions, Liao Ning Province	2020

TA Experience

• Functional Analysis III	BNU
• <i>Organize Q & A and exercise classes for over 100 students.</i>	<i>Sep. 2025 – Dec. 2025</i>
• Mathematical Analysis I (Outstanding TA in 2024)	BNU
• <i>Organize Q & A and exercise classes for over 30 students.</i>	<i>Sep. 2024 – Dec. 2024</i>
• Mathematical Analysis III	BNU
• <i>Organize Q & A and exercise classes for over 110 students.</i>	<i>Sep. 2023 – Dec. 2023</i>

Conference & Talks

• Harmonic Analysis and Its Applications	Beijing, China
• <i>Participant</i>	<i>Nov. 2025</i>
• BNU Harmonic Analysis Forum	Virtual Online
• <i>Participant</i>	<i>2024-2025</i>
• Harmonic Analysis and Its Applications	Beijing, China
• <i>Participant</i>	<i>Oct. 2023</i>
• Beijing Workshop on Harmonic Analysis	Beijing, China
• <i>Participant</i>	<i>Aug. 2023</i>

Skills

- **Programming:** Proficient in Matlab, Python, Markdown and L^AT_EX. Familiar with C++, R and SQL.
- **Languages:** English (TOEFL - 100); Mandarin (native).
- **Sports:** Volleyball.

Relevant Coursework

• Functional Analysis	• Graph Theory	• Numerical Algebra
• Fourier Analysis	• Numerical Approximation	• Time Series Analysis